

### ***Multi – level Linked List***

***–Destroy List(Main List and Child)***

***–Time Complexity Calculation using Akrabazzi Method***

$$2T\left(\frac{N-1}{2}\right) + C, \text{ Base Case: } T(0) = 1.$$

### **Solution**

***a = 2 and b = 2 ∴***

***$\frac{a}{b^p} = 1$ , only when  $2 = 2^p$  or  $\frac{2}{2^p} = 1$ , when  $p = 1$ .***

***Therefore, as per Akra – Bazzi theorem ,  
the solution can be given as follows:***

$$2T\left(\frac{N-1}{2}\right) + C$$

***u = n – 1 , f(n) = C and p = 1***

$$\Theta\left(n^p\left(1 + \int_1^n \frac{f(n)}{n^{p+1}} dn\right)\right)$$

$$\Rightarrow \Theta\left(n^1\left(1 + \int_1^n \frac{c}{n^{1+1}} dn\right)\right)$$

$$\Rightarrow \Theta \left( n \left( 1 + \int_1^n \frac{c}{n^2} dn \right) \right)$$

$$\Rightarrow \Theta \left( n \left( 1 + c \left( \int_1^n \frac{1}{n^2} \right) dn \right) \right)$$

$$\Rightarrow \Theta \left( n \left( 1 + c \left( \int_1^n n^{-2} \right) dn \right) \right)$$

*Lets solve  $\int_1^n n^{-2} dn$  this part first:*

$$\text{We know: } \int x^a = \frac{x^{a+1}}{a+1}$$

$$\int_1^n n^{-2} \Rightarrow [n^{-2}]_1^n \Rightarrow \left[ \frac{n^{-2+1}}{-2+1} \right]_1^n \Rightarrow \left[ \frac{n^{-1}}{-1} \right]_1^n \Rightarrow \left[ \frac{1}{-1} \times n^{-1} \right]_1^n$$

$$\Rightarrow [-1 \times n^{-1}]_1^n \Rightarrow \left[ -1 \times \frac{1}{n} \right]_1^n \Rightarrow \left[ -\frac{1}{n} \right]_1^n$$

***Now compute boundaries of  $-\frac{1}{n}$ :***

$$\begin{aligned}\int_b^a f(x) d(x) &= [f(x) d(x)]_b^a = F(b) - F(a) \\ &= \lim_{x \rightarrow b^-} F(x) - \lim_{x \rightarrow a^+} F(x)\end{aligned}$$

***Here,  $x = n$ , Therefore,***

$$\begin{aligned}\lim_{n \rightarrow n^-} F(x) &= \lim_{n \rightarrow n^-} \left(-\frac{1}{n}\right), \text{ Since } \lim_{x \rightarrow x} f(x) = f(x) \\ \therefore \lim_{n \rightarrow n^-} \left(-\frac{1}{n}\right) &= -\frac{1}{n}\end{aligned}$$

$$\begin{aligned}\lim_{n \rightarrow 1^+} \left(-\frac{1}{n}\right) &= \text{plugging 1 in place of } n \text{ we get:} \\ &= -\frac{1}{1} = -1\end{aligned}$$

$$\therefore F(b) - F(a)$$

$$\Rightarrow -\frac{1}{n} - (-1)$$

$$\Rightarrow -\frac{1}{n} + 1$$

*Applying it in :  $\Theta\left(n\left(1 + c\left(\int_1^n n^{-2}\right) dn\right)\right)$  , we get:*

$$= \Theta\left(n\left(1 + c\left(-\frac{1}{n} + 1\right)\right)\right)$$

$$= \Theta\left(n\left(1 + \left(-\frac{c}{n} + c\right)\right)\right)$$

$$= \Theta\left(n\left(1 - \frac{c}{n} + c\right)\right)$$

$$= \Theta\left(n \times 1 - n \times \frac{c}{n} + nc\right)$$

$$= \Theta(n - c + nc)$$

$$= \Theta(n + nc - c)$$

$$= \Theta(n)$$

*Hence, through Akrabazzi Method it is proved that :*

***Time Complexity :  $\Theta(n)$***

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